

Learning Unit 2: Enterprise and Functional Business Information Systems

Learning Objectives: • Identify key concepts of enterprise and functional business information system; • Describe the major types of business information systems and their functions; • Explain how information systems support different organisational areas.	My notes
Material used for this learning unit: • The content of this learning unit is aligned to chapter 6 of Business Information Systems by Bocij et al. (2019)	
How to prepare for this learning unit: • Work through the contents in this learning unit and complete the activities at the end of the learning unit.	

1. Introduction

The value of Business Information Systems (BIS) for an organisation depends on how hardware, software, and network technologies are used to support the organisation's objectives. This is achieved through the implementation of specific business applications that aid various organisational processes and functions. BIS can be classified into operational and management systems, which can be deployed as enterprise or functional business systems.

This learning unit focuses on enterprise systems and the main types of enterprise systems, including Enterprise Resource Planning (ERP), Supply Chain Management (SCM), Customer Relationship Management (CRM), and Supplier Relationship Management (SRM). It also examines operational information systems, such as transaction processing systems, manufacturing systems, and office automation systems, as well as management information systems, including decision support systems for tactical and strategic planning.

Furthermore, this learning unit briefly discusses the use of different software types in various functional areas of a business, such as finance, accounting, human resources, and marketing.

2. Enterprise systems

Enterprise systems aim to support an organisation's business processes across functional boundaries (Bocij et al., 2019). They utilise Internet technology to integrate information internally and with external stakeholders like customers, suppliers, and partners. The main types of enterprise systems are enterprise resource planning (ERP) for internal production, distribution, and financial processes; customer relationship management (CRM) for marketing and sales processes; supply chain management (SCM) for material, information, and customer flow; and supplier relationship management (SRM) for sourcing, purchasing, and warehousing (Bocij et al., 2019). Other types of ES include product lifecycle management (PLM), financial management, and human capital management.

2.1. *Characteristics of enterprise systems*

Enterprise systems are known for their cross-functional process view of an organisation (Bocij et al., 2019). They include defined business process designs, or blueprints, for functions like procurement, production, and fulfilment (Bocij et al., 2019). Another key feature of enterprise systems is their centralised database structure, which allows for seamless data integration across the organisation (Bocij et al., 2019).

2.2. *The benefits and main disadvantages of the enterprise systems*

According to Bocij et al. (2019), the following are the benefits of the enterprise systems:

- Integration of organisational processes leading to increased efficiency and improved quality of customer service;
- The defined business process blueprints in the Enterprise Systems can serve as a template for a BPM initiative;

- Improved information sharing within the organisation resulting from module integration, leading to better decision-making and increased organisational agility;
- Simplified support and maintenance through a single supplier, eliminating the need to manage multiple legacy systems;
- Adoption of best-of-breed enterprise systems solutions with proven functionality in specific areas like procurement systems.

The following are the main disadvantages of the enterprise systems (Bocij et al., 2019):

- High costs charged by suppliers for a large, complex system;
- Implementing major organisational changes requiring expensive planning, training, and development to transform systems and processes;
- Adapting to standardised ERP blueprints, like SAP's 1000+ predefined activities, can reduce competitive advantage and hinder process innovation.

2.3 *Types of enterprise systems*

2.3.1 Enterprise Resource Planning (ERP) systems

Enterprise Resource Planning (ERP) systems provide function for managing business areas such as production, distribution, sales, finance and human resource management (Bocij et al., 2019).

Enterprise Resource Planning (ERP) systems are designed to support various processes within an organisation, including procurement, product development, manufacturing, and sales (Bocij et al., 2019). They can be expanded to include additional enterprise systems like supply chain management (SCM) systems, which manage processes such as warehousing, transportation, and supply network collaboration (Bocij et al., 2019). Customer relationship management (CRM) systems focus on marketing, sales, and service, while supplier relationship management (SRM) systems handle sourcing, contract management, supplier collaboration, and procurement (Bocij et al., 2019). These different types of enterprise systems play an important role in enhancing organisational efficiency and productivity.

2.3.2 Supply chain management (SCM) systems

The supply chain is a series of activities that move materials from suppliers to customers through the organisation (Bocij et al., 2019). Each product or service has its own supply chain, involving multiple organisations in processing, transportation, warehousing, and retail (Bocij et al., 2019).

2.3.3 Customer relationship management (CRM)

CRM systems are centred around a database, and when employees and customers access this database through a website, it is known as e-CRM (Bocij et al., 2019). According to Bocij et al. (2019), common applications integrated into a CRM system may include:

1. Customer data collection: This involves gathering personal details such as age, gender, and contact address, as well as recording purchase transactions including location, date, time, quantity, and price. This information can help call centre staff enhance and personalise their services for individual customers.

2. Customer data analysis: The collected data enables the categorisation and targeting of customers based on criteria set by the company. This information can enhance the effectiveness of marketing campaigns.

3. Salesforce automation: CRM facilitates the entire sales cycle, from lead generation to closing the sale and providing after-sales service.

2.3.4 Supplier relationship management (SRM)

SRM systems facilitate all activities involved in obtaining items from a supplier, which includes not only procurement but also inbound logistics such as transportation and warehousing (Bocij et al., 2019).

3. Operations Information Systems

Operations information systems (OIS) are essential for the daily operations of a business (Bocij et al., 2019). They play a crucial role in the organisation and are often referred to as mission-critical or strategic information systems. According to Bocij et al. (2019), there are three main types of operational systems:

1. Transaction processing systems (TPS) are used to facilitate the exchange of information and funds between a company and external parties like customers, suppliers, and distributors (Bocij et al., 2019). Additionally, they help organisations manage the frequent external and internal transactions such as orders for goods and services that serve the operational level of the organisation (Bocij et al., 20219).

A TPS can help an organisation gain a competitive edge in various ways (The IIE, 2019):

- 1. Operational Efficiency:** TPS streamlines tasks, reducing manual efforts and enhancing efficiency.
- 2. Cost Reduction:** By automating tasks, TPS reduces the need for extensive human involvement, cutting personnel and operational costs.
- 3. Real-time Decision Making:** TPS processes data in real-time, enabling timely and informed decisions.
- 4. Accurate Information:** TPS ensures accurate data processing, providing reliable and up-to-date information for sound decision-making.
- 5. Improved Customer Service:** TPS enhances customer service by enabling quick access to information and efficient processes.

2. Office automation systems (OAS) are essential for managing administrative functions in office environments, particularly in service-based industries (Bocij et al., 2019). These information systems are intended to increase the productivity of office workers (Bocij et al., 2019).

Office Automation Systems (OAS) aim to enhance the efficiency of office workers by providing tools such as groupware, workflow systems, word processors, and spreadsheets (Bocij et al., 2019). Examples include groupware, workflow, and general-purpose applications like word processors and spreadsheets. Mission-critical applications such as groupware and workflow are essential for supporting the internal processes of e-businesses (Bocij et al., 2019).

According to Laudon and Laudon (2018), OAS play **three key organisational roles**:

1. Coordinating and overseeing the tasks of local, professional, and information workers within the organisation;
2. Connecting the work carried out across different levels and departments of the organisation.
3. Establishing links between the organisation and its external environment, including clients and suppliers. For instance, contacting an organisation means reaching its office.

3. Process control systems play a crucial role in manufacturing industries by overseeing the production of goods (Bocij et al., 2019). These systems provide real-time information to employees directly involved in a particular activity. These Information systems can support three main types of production facilities (Bocij et al., 2019):

1. **Repetitive**: These are production-line systems that produce standard products, similar to packaged software.
2. **Job shop**: These facilities produce individualised products for specific customers, similar to bespoke software.
3. **Batch**: These facilities produce batches of identical products before changing the production setup for the next batch.

4. Management Information Systems

Management information systems are defined as systems that provide feedback on organisational activities and support managerial decision-making (Bocij et al., 2019). There are three main types of management information systems, namely, decision support systems (DSS), information reporting systems (IRDS), and executive strategic systems (EIS), (Bocij et al., 2019):

4.1 Decision Support Systems (DSS) offer information and models to aid tactical and strategic decision-making (Bocij et al., 2019). Decision Support Systems (DSS) are typically created by end-users and are specific to departments rather than being corporate-wide systems (Bocij et al., 2019). DSS are commonly used for ad-hoc queries rather than routine reporting (Bocij et al., 2019). They are often used for marketing purposes, including:

- Predicting sales by analysing geodemographic data;
- Improving distribution networks by selecting optimal retail locations through modelling;
- Enhancing product assortments.

The following key objectives for Decision Support Systems (DSS) are proposed (The IIE, 2019; Bocij et al., 2019):

1. The DSS should assist in decision-making, especially for semi-structured and unstructured decisions;
2. The DSS should cater to decision-making at all levels of management, including operational, tactical, and strategic levels;
3. The DSS should support all stages of the decision-making process;
4. The DSS should be user-friendly.

4.2 Information reporting systems (IRS) generate pre-defined reports for daily decision-making (Bocij et al., 2019). There are two main types of reports generated by these systems (Bocij et al., 2019):

- 1. Periodic reports:** These are predefined reports needed by decision-makers at regular intervals, such as monthly financial statements and weekly sales analyses.
- 2. Exception reports:** These reports are generated automatically when performance thresholds are breached or manually when timely or occasional information is needed.

4.3 Executive information systems (EIS) offer senior managers a platform to analyse, compare, and identify trends to guide the strategic direction of a company (Bocij et al., 2019). A typical example would be to monitor the organisation's performance against changes in the external environment such as competitor actions.

Some important features of EIS are as follows (Bocij et al., 2019):

- Providing summary information for monitoring business performance;

- Primarily used for strategic decision-making, with potential for tactical decision support;
- Offering a drill-down capability for detailed information retrieval;
- Including analysis tools for data interpretation;
- Integration with other systems for problem-solving and daily operations management;
- Consolidating data from various sources, both internal and external;
- Tailoring design to suit non-computer-savvy managers.

4.4 Functional or Departmental information systems

Information systems can be designed to support different functional areas within an organisation (Bidgoli, 2022: The IIE, 2019). A functional area in an organisation consists of a group of people who perform similar tasks. Examples of functional areas in an organisation include manufacturing, IT, human resources, logistics, etc.

Below are examples of how information systems can support different functional areas within an organisation (Bidgoli, 2022: The IIE, 2019):

4.4.1 Human Resource Information System (HRIS)

An HRIS provides support to the human resources functional area of an organisation, helping its members perform their tasks more efficiently (Bidgoli, 2022).

An HRIS can assist the members of the human resources functional area in making the following decisions (Bidgoli, 2022):

- Choosing the best job candidate;
- Scheduling and assessing employees;
- Predicting future staffing needs;
- Generating reports and statistics on employee demographics.

4.4.2 Logistics Information Systems (LIS)

The Logistics Information System was developed with the aim of reducing the costs of processes involved in the logistics functional area of an organisation (Bidgoli, 2022).

Decisions supported by the LIS include (Bidgoli, 2022):

- Improving routing and delivery schedules;
- Selecting the best modes of transportation;
- Improving budgeting for transportation;
- Improving shipment planning.

4.4.3 Manufacturing Information Systems (MFIS)

The MFIS was developed to reduce manufacturing costs and improve product quality (Bidgoli, 2022).

Decisions supported by the MFIS include (Bidgoli, 2022):

- Ordering decisions;
- Product cost calculations;
- Space utilisation;
- Decisions related to tenders;
- Decisions in relation to price changes and discounts.

4.4.4 Financial Information Systems (FIS)

An FIS provides timely information to the members of an organisation who are responsible for making financial decisions (Bidgoli, 2022).

An FIS can assist in making decisions related to (Bidgoli, 2022):

- Budget allocations;
- Capital investment risk management;
- The monitoring of cost trends;
- The managing of cash flows;
- Determining portfolio structures.

